



DIGITAL LOGIC DESIGN (DLD)

Chapter 3: Gate Level Minimization

Week 4: Lecture 2 part 1

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Reference Readings and Acknowledgements

Chapter 2: Digital Design with Introduction to
Verilog HDL M. Morris Mano, Michael D.
Ciletti, 5th Edition

Contents

- Introduction
- The Map Method
- Four-Variable Map
- Five-Variable Map
- Product-of-Sums Simplification
- Don't-Care Conditions
- NAND and NOR Implementation
- Other Two-Level Implementations
- Exclusive-OR Function

Introduction

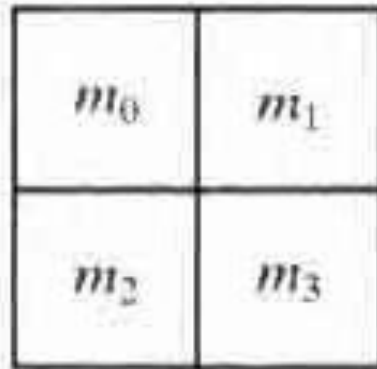
- *Gate-level minimization*

- finding an optimal gate-level implementation of the Boolean function.
- Difficult to execute by manual methods when the logic has more than a few inputs.

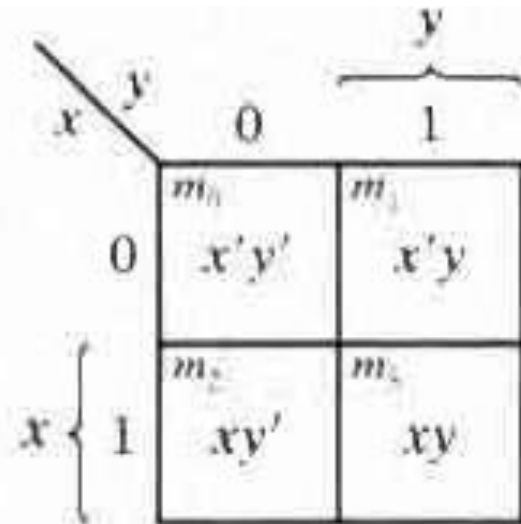
The Map Method

- A simple, straightforward procedure for minimizing Boolean functions.
 - a pictorial form of a truth table.
 - known as the *Karnaugh map* or *K-map*.
 - Diagram is made up of squares, with each square representing one minterm of the function
 - represents a visual diagram of all possible ways a function may be expressed in standard form.
 - By recognizing various patterns. the user can derive alternative algebraic expressions for the same function.
 - from which the simplest can be selected.
 - Simplified expression can be SOP or POS
 - with a minimum number of terms and with the smallest possible number of literals in each term.
 - Simplest expression is not necessarily be unique.

Two-Variable Map



(a)



(b)

$$m_1 + m_2 + m_3 = x'y + xy' + xy = x + y$$

Three-Variable Map

m_0	m_1	m_3	m_2
m_4	m_5	m_7	m_6

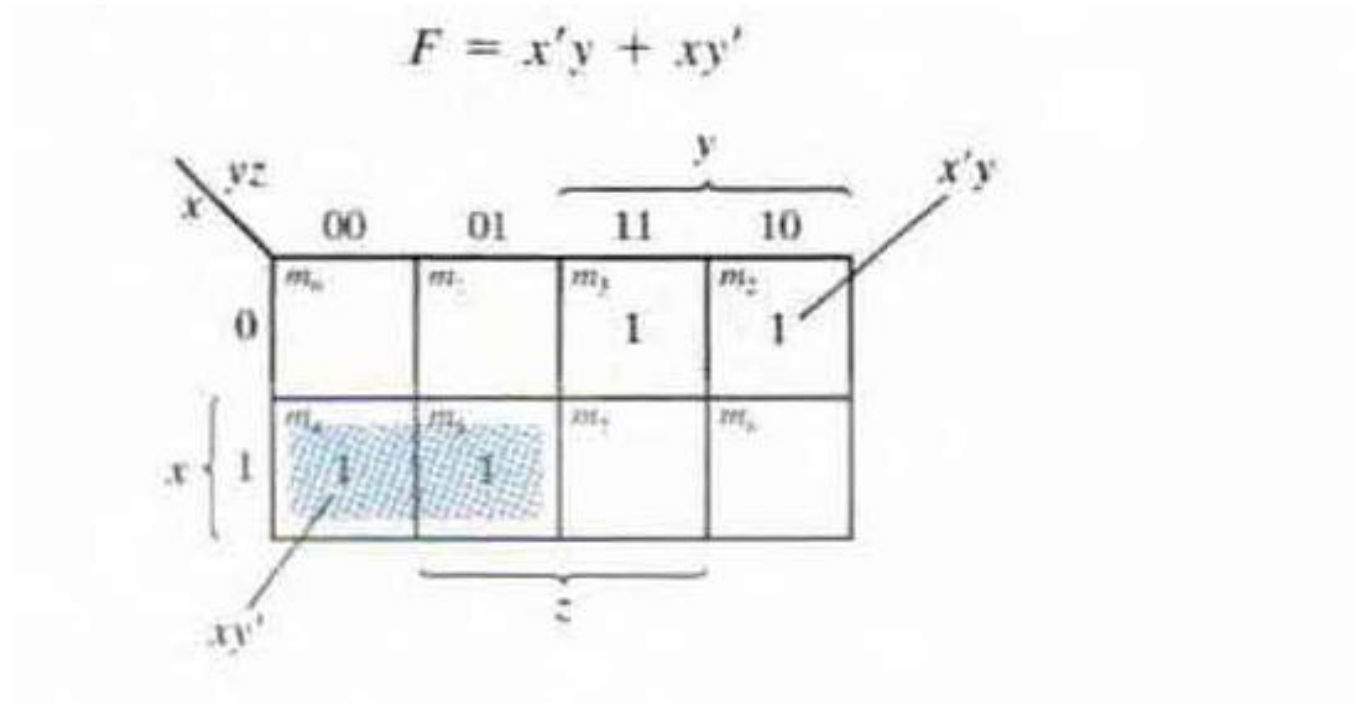
(a)

		y			
		00	01	11	10
x	0	m_0 $x'y'z'$	m_1 $x'y'z$	m_3 $x'yz$	m_2 $x'yz'$
	1	m_4 $xy'z'$	m_5 $xy'z$	m_7 xyz	m_6 xyz'
		z			

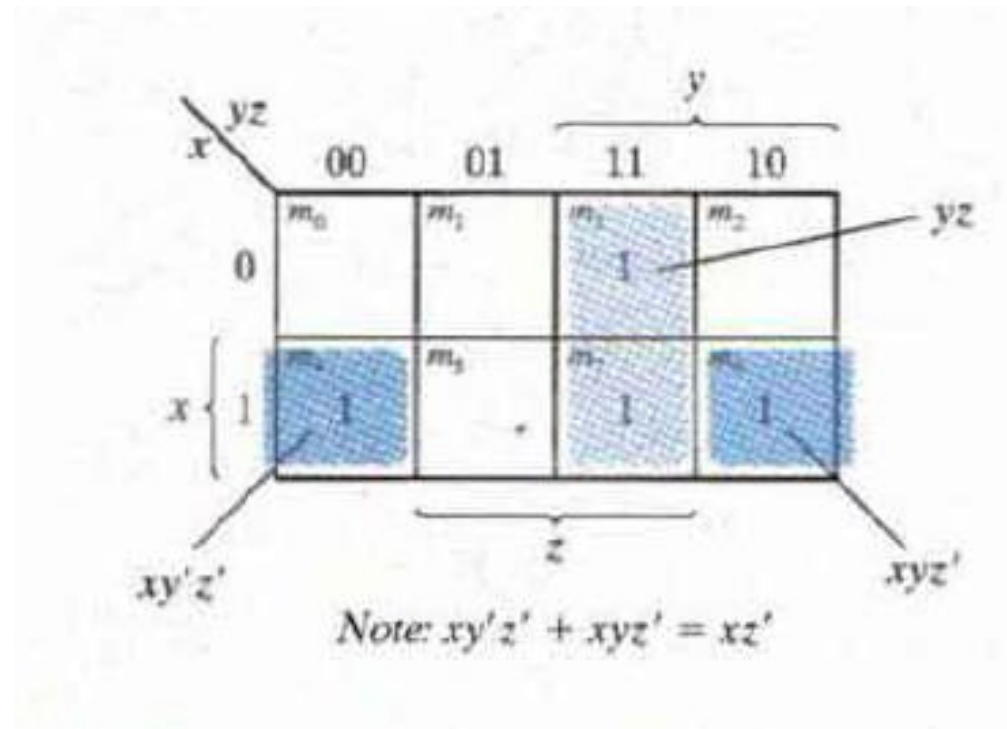
(b)

FIGURE 3.3
Three-variable map

Example 3.1 $F(x, y, z) = \sum (2, 3, 4, 5)$



Example 3.2 $F(x, y, z) = \sum (3, 4, 6, 7)$



Example 3.3 $F(x, y, z) = \sum (0, 2, 4, 5, 6)$

